RESILIENT CLINICAL OUTCOME PREDICTION

FROM NOISY MEDICAL RECORDS

Predicting Hospital Readmission in Diabetic Patients Using Robust Machine Learning Pipelines

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01 ABSTRACT

Hospital readmission within 30 days is a critical quality indicator and an **avoidable** cost burden in healthcare systems. This project applies machine learning to the **UCI Diabetes 130-US Hospitals Dataset** — a real-world, noisy electronic health record (EHR) spanning 10 years (1999–2008) — to predict patient readmission outcomes.

We built an end-to-end ML pipeline addressing **severe class imbalance, high data missingness (up to 97%), and noisy categorical features**. Three classifiers were rigorously evaluated: Logistic Regression, Decision

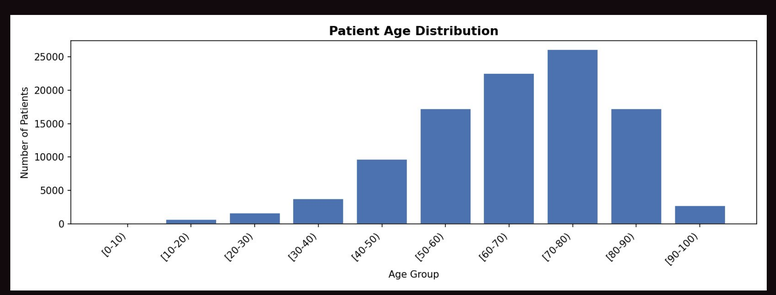
02 DATASET OVERVIEW

101,766	Patient Records
50	Clinical Features
10 yrs	Data Span (1999–2008)
3	Target Classes
97%	Weight Data Missing
[70-80]	Dominant Age Group

03 METHODOLOGY

Data Ingestion	Raw EHR CSV loaded & audited
Data Cleaning	Dropped cols >40% missing; mode imputation
Feature Eng.	Label encoding, binary targets, age groups
SMOTE	Synthetic oversampling for minority class
Model Training	LR · DT · RF with cross-validation
Evaluation	AUC-ROC · F1 · Precision · Recall · Acc

04 AGE DISTRIBUTION



05 KEY FINDINGS

Elderly Population Dominance

The 70–80 age group formed the largest cohort (~26,000 patients), driving strong age-based feature importance across all models.

Medications → Hospital Stay

num_medications correlated most strongly with time_in_hospital (r=0.47), identifying complex patients needing enhanced post-discharge follow-up.

Severe Data Quality Issues

Weight (97% missing), medical_specialty (49%), and payer_code (40%) were dropped. Mode imputation applied to race (~2% missing).

Random Forest Wins on Robustness

RF tied LR on metrics but demonstrated superior noise resilience and consistent performance across all data fractions — recommended for production.

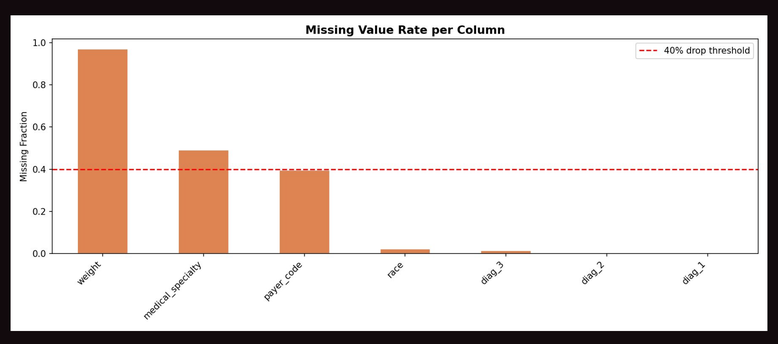
Class Imbalance Persists

Despite SMOTE, AUC-ROC remained ~0.59 — highlighting the inherent difficulty of predicting early readmission from EHR data alone.

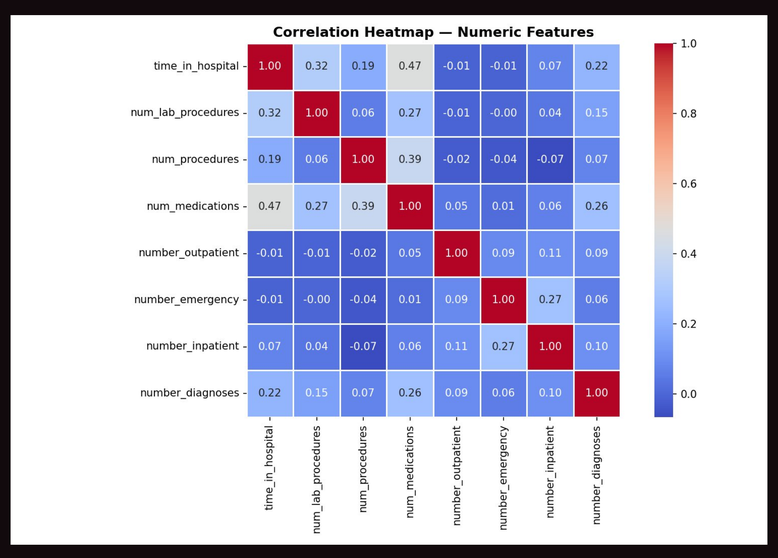
Near-Linear Scalability

Training time grew from 7.74s to 9.76s as data 4×'d (22K → 91K rows), confirming suitability for hospital-scale deployment.

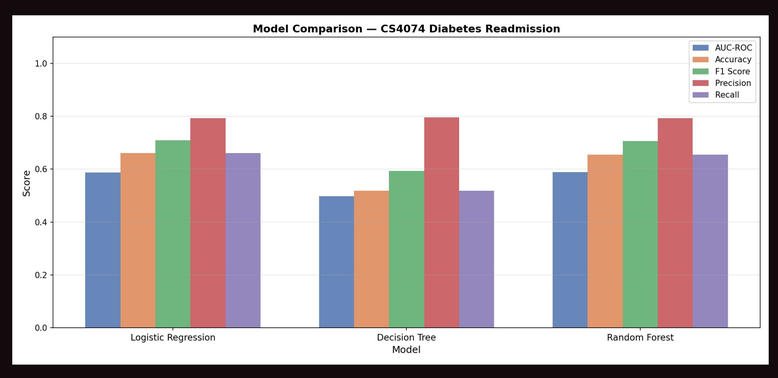
06 MISSING VALUE ANALYSIS



07 CORRELATION HEATMAP



08 MODEL COMPARISON



09 PERFORMANCE METRICS

Model	AUC	Acc	F1	Prec	Rec	
Logistic Reg.	0.59	66%	71%	79%	66%	★
Decision Tree	0.50	52%	59%	80%	52%	
Random Forest	0.59	66%	71%	79%	66%	★

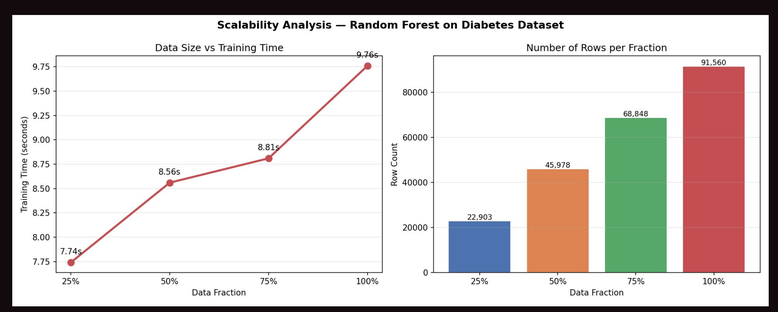
AUC-ROC: 0.59

Accuracy: 66%

F1 Score: 71%

Precision: 79%

10 SCALABILITY ANALYSIS



11 CONCLUSIONS & QR

Logistic Regression and Random Forest tied on ke metrics (Acc=66%, F1=71%, Prec=79%), significantl outperforming Decision Tree (Acc=52%). Randor Forest is preferred for production due to superior nois tolerance and linear scalability.

Future work: XGBoost / LightGBM ensembles, dee learning on raw text notes, and federated learning across hospital networks for privacy-preserving model training.

SCAN FOR REPO

Dataset: UCI ML Repository — Diabetes 130-US Hospitals (1999-2008)

Strack et al. (2014) · 101,766 encounters · 50 features

Best: LR & RF

Accuracy: 66% · F1: 71%

AUC-ROC: 0.59